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```

TING
thank you for buying EP-2350 TING, a full instruction and warranty terms can be
found on teenage engineering's website https://teenage.engineering

using TING
power it by inserting 2 x AAA batteries or attaching to a USB power source.
push the handle to start. you must also push the handle to enable the microphone.

on batteries:
  * 5 min after releasing the handle the unit goes into power save mode
  * 20 min after releasing the handle the unit turns off
  * if the battery voltage is too low an LED will blink
    and the unit will not turn on
  * remove the batteries if the unit is stored

on usb power:
  * the unit will stay powered until you pull the cable
  * the batteries are not used if there is usb power
  * the batteries are not charged

WARNING: loud sound levels leads to hearing loss. TING is designed to connect
to the RIDDIM or an audio mixer, not directly to headphones.

signal level out is 2 VRMS and this can be very loud connecting directly to
headphones, under the lid there is a round green turn to adjust the volume.

the orange button selects between clean voice and 4 effect presets
some of the effect parameters are modulated by the position of the handle,
some are effected by if you shake the unit.

    [clean, echo, echo+spring, pixie, robot]

the green button selects between 4 preset samples. if you want to
you can change these to your own samples.

    [siren, alarm, gunshot, monkey boy]

the white button plays the selected sample.

pushing the handle

the position of the handle can change the amount of an effect.

shaking the unit

shaking the unit temporarily modifies the sound of the effect,
how is preset dependent.

upgrading the firmware

first you will need a firmware uf2 file. if there is a new firmware this can
be downloaded from teenage engineering's website.
remove the lower lid and attach a usb-c cable to your computer,
hold in the handle and then double click the small button above the usb port.
a drive called TING BOOT should appear in your computer containing the files
INDEX.HTM and INFO_UF2.TXT, drag and drop the firmware file to this drive
and wait until transferred and the unit restarts. do not release the handle
until the the operation is finished.

changing the samples

overriding a preset sample with your own sample is done by placing files called
1.wav, 2.wav, 3.wav and 4.wav on the TING DISK. only wav format is allowed and
the total file size is about 1 MB. the samples can be mono or stereo, 8/16/24-bit
or 32-bit floating point, with a frequency up to 96 kHz. the samples are not
available until the unit is restarted, push the button over the usb-c port
turn it off and push the handle to start again.

config.json

a file called config.json overrides the presets and selects what samples to play.
json is a human-readable data exchange format that that TING can interpret.
please look at the example syntax and you will be able to write your own.
some effects like delay, reverb, harmony and ssb should only be used ones per
effect chain. parameters not initialized will have a default value. if your
config file is so extreme that the unit will not start, hold green + white
button when starting and then fix the file.

example
~~~
{
  "name": "We count from zero",
  "samples": [
    { "pos": 1, "file": "samples/whistle1.wav","playmode": "oneshot" },
    { "pos": 0, "file": "live1/loop.wav","playmode": "startstop" },
    { "file": "horn.wav","playmode": "hold" },
    { "file": "live1/shottis.wav","playmode": "oneshot" }
  ],
  "presets": [
    {
      "pos": 0,
      "list": [
        { "effect": "HARMONY", "pitch": 2.0, "BUS": 2   },
        { "effect": "REVERB", "time": 0.1, "dry-level": 1.0 },
        { "effect": "DELAY","time": 0.5, "dry-level": 0.0, "echo": 0.5, "BUS": 1 },
        { "effect": "SAMPLE", "speed": 1.0 }
      ],
      "handle": { "row": 1, "param": "time","depth": 0.6 },
      "shake": { "row": 2, "param": "echo", "depth": 1.0 },
      "lfo": { "row": 3, "param": "echo", "depth": 1.0,
               "mpy": 1.0, "shape": "random", "phase": 0, "speed", 4.0 },
      "trigger": { "row": 3 }

    },
    {
      "pos": 2,
      "list": [
        { "effect": "HARMONY", "pitch": 2.0 },
        { "effect": "SAMPLE", "speed": 2.0 },
        { "effect": "REVERB", "time": 1.0, "spring": 0.5 },
        { "effect": "DELAY", "time": 0.1, "echo": 0.5 }
      ],
      "handle": { "row": 2, "param": "pitch", "depth": -0.2 },
      "trigger": { "row": 1 }

    }
  ]
}
~~~

effects
```
    BALANCE:
        balance             [0.0, 1.0]

    DELAY:
        time                [0.0, 1.2]
        lowpass-cutoff      [0.0, 1.0]
        highpass-cutoff     [0.0, 1.0]
        wet-level           [0.0, 1.0]
        dry-level           [0.0, 1.0]
        echo                [0.0, 1.0]
        cross-feed          [0.0, 1.0]
        balance             [0.0, 1.0]

    DIST:
        amount              [0.0, 40.0]
        highpass-cutoff     [0.0, 1.0]
        lowpass-cutoff      [0.0, 1.0]
        mix                 [0.0, 1.0]

    HARMONY:
        dry-level           [0.0, 1.0]
        pitch               [0.50, 2.0]

    HIGHPASS:
        cutoff              [0.0, 1.0]

    LOWPASS:
        cutoff              [0.0, 1.0]

    SAMPLE:
        speed               [0.0, 4.0]
        pitch               [-24.0, 24.0]
        level               [0.0, 1.0]
        balance             [0.0, 1.0]

    REVERB:
        dry-level           [0.0, 1.0]
        wet-level           [0.0, 1.0]
        time                [0.0, 1.0]
        spring-mix          [0.0, 1.0]
        highpass-cutoff     [0.0, 1.0]

    RING:
        frequency           [0.0, 20000.0]
        mix                 [0.0, 1.0]

    SSB:
        frequency           [-20000.0, 20000.0]
```

LFO shapes

    [sine, square, sawtooth, random]

Licenses